

Application of Hilbert-Huang Transform for the Study of Motor Imagery Tasks

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Abstract— A motor based Brain-Computer Interface (BCI) translates the subject's motor intention into a control signal by means of the method which extracts characteristic feature from EEG recorded from the scalp. In this paper, the EEG signal recorded during three motor imagery tasks, which were imagination of left hand, right hand and foot movements, was investigated. A novel method named Hilbert Huang transform (HHT) is introduced to extract the feature from signal. Firstly, raw signal is decomposed using Empirical Mode Decomposition (EMD). And then, several Intrinsic Mode Functions (IMF) are gained. For further study, the IMFs whose main frequency is higher than 5 Hz are selected. Secondly, based on the IMFs selected above, hilbert spectrum is calculated. In each motor imagery task, local instantaneous energies, within specific frequency band of electrode C3 and C4, are selected as the features. A three-layer BP Neural Network classifier is structured for pattern classification. The classification results show that HHT can be used in EEG-based BCI research as a method to analysis non-linear and non-stationary signal.

I. INTRODUCTION

BRAIN-COMPUTER interface is an emergent multidisciplinary technology which allows a brain to control a computer directly – without depending on the brains normal output pathways of peripheral nerves and muscles.[1] It is suitable for people that are incapable of any motor functions, such as people with amyotrophic lateral sclerosis (ALS), high-level spinal cord injure or brain stem stroke. Among brain signal measures, electrical signal recorded from scalp, termed electroencephalogram (EEG) is widely investigated. EEG has downsides like low signal-to-noise ratio, low spatial resolution, but the advantages such as non-invasive, low cost and high temporal resolution made it popular in BCI research.

The key problem in non-invasive BCI research is how to extract the feature which exactly represents the human intent from the EEG. Features related with different electrophysiological background are concerned. The features include the amplitude of mu or beta rhythm (the subject can

control the amplitude of some brain waves after some training), the potentials when subject received some external stimuli (event-related potentials), energy change at particular channel when subject imagined movement of his limb. In present paper the EEG signal recorded during motor imagination is investigated.

According to the neurophysiological observation, imagination of left or right hand movement can attenuate the mu and beta rhythm activity, which is termed event-related desynchronization (ERD). On the contrary, the activity which can enhance the two types of brain wave is termed event-related synchronization (ERS) [2]. In this paper, when the subjects are performing different motor imagination tasks, the difference in energy is extracted from particular rhythm as characteristic feature by the method of HHT.

II. EXPERIMENT AND DATA ACQUISITION

A. Subjects

Three right-handed volunteers (two males and one female, 27~29 years old) participated in this study. All of these subjects had no experience with BCI experiments.

B. Experiments

The subjects sat in an armchair and stared at the computer monitor placed approximately 70 centimeters away from the subject at eye level. Meanwhile, they were asked to keep their arms and hands relax and comfortable and avoid eye movements during the recordings.

The subjects were told to imagine left hand, right hand or foot movement following the cue emerged on the screen. Each trail started with a blank screen lasting for 2 seconds. At second 2, a fixation cross appeared at the center of the monitor. At second 3, the cross disappeared and at the same time an arrow emerged at the center of the monitor, which was pointing left, right, or upward. The subjects were instructed to imagine a movement of the left hand, right hand or the foot, depending on the direction of the arrow. The arrow vanished after 2.5 seconds, then a letter 'S' showed in the monitor which means stop imagine movement. The subject can relax during this period which would last for 2 seconds. The whole experiment paradigm is shown in Fig.1. One session contains twenty times of each movements. Every subject performed four sessions in one day.

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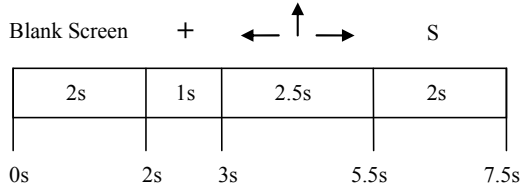


Fig. 1 Timing scheme of one trail

C. Data Recording and Preprocessing

EEG was recorded using Ag/AgCl electrodes in a 64 channel modified quick cap (NeuroScan, Inc). The electrodes are evenly spaced and symmetrically covered the scalp from nasion toinion and from left to right ear, according to the 10-20 system. The reference electrodes were positioned on left and right mastoids. Horizontal EOG recoding electrodes were positioned at the outer canthi of both eyes and vertical EOG recording electrodes were placed above and below the left eye. All signals, including EEG and EOG were sampled at 250 Hz.

The original EEG signals were preprocessed using Neuro Scan 4 software system. During this process, two things were done. One is to remove the disturbances caused by EOG; the other is bandpass filter between 0.1Hz and 40Hz.

Eleven EEG channels which placed in the primary motor area were chosen to investigate in this study. The electrodes were C3, C4, CP3, CP4, FC3, FC4, C1, C5, C2, C6 and Cz.

III. METHOD

A. Hilbert-Huang transformation

Hilbert-Huang transformation, first proposed by Huang et al [3], is a method to analyze non-linear and non-stationary signals. Its basis of expansion is adaptive, so that it can produce physically meaningful representations of data from non-linear and non-stationary processes. In present day, HHT has been applied to many scientific research areas, such as biomedical signals processing [4][5], geophysics [6], image processing [7], mechanical faults diagnosis [8] and so on. Hilbert-Huang transformation can be divided into two parts, Empirical mode decomposition and Hilbert-Huang transform. We will introduce this method in detail.

1. Empirical mode decomposition method.

Before introduce the EMD, the definition of intrinsic mode function (IMF) is given. According Huang's [3] definition IMF as a class of function that should satisfy two conditions:

- 1) The number of extrema and the number of zero-crossings must be either equal or differ at most by one;
- 2) At any point, the mean value of the envelope defined by the local maxima and the envelope defined by local minima is zero.

Given a signal $x(t)$, the algorithm of EMD can be summarized as follows[3].

- 1) Identify all extrema (maxima and minima) of $x(t)$.
- 2) Generate of the upper and lower envelopes via interpolation among all the maxima and minima.

3) Compute the average of the two envelopes, the average defined as $m(t)$.

4) Subtraction of $m(t)$ from the data to obtain a IMF candidate $c(t)=x(t)-m(t)$.

5) If $c(t)$ does not satisfy the previously defined properties of IMF, replace $x(t)$ with $c(t)$ and repeat the Procedure from step 1; if $c(t)$ satisfies the previously defined properties of IMF, take $c(t)$ as a IMF, and evaluate the residue $r(t)=x(t)-c(t)$.

6) Repeat the procedure for step 1 to 5 by sifting the residual signal. When the residue $r(t)$ satisfy some stopping criterion, the process ends.

Using EMD method, the signal $x(t)$ is decomposed to a residue $r_n(t)$ and a collection of n IMFs, as $c_i(t)$ ($i=1,2,\dots,n$). $r_n(t)$ is a mean trend or a constant. $c_i(t)$ is the IMF, which sorted in descending order of frequency. $c_1(t)$ associated with the locally highest frequency and $c_n(t)$ related with the lowest frequency. The original $x(t)$ can be exactly reconstructed by a linear addition:

$$x(t) = \sum_{i=1}^n c_i(t) + r_n(t) \quad (1)$$

Each IMF is a signal with zero mean. It can be regard as a signal produced by a specific oscillation source. An IMF represents a simple oscillatory mode as a counterpart to the simple harmonic function, but it is much more general. Instead of constant amplitude and frequency, as a simple harmonic component, the IMF has a variable amplitude and frequency as functions of time.

2. The Hilbert spectral analysis.

Having obtained the IMFs using EMD method, one applies the Hilbert transform to each IMF component. Amplitude $\alpha_i(t)$ and phase $\theta_i(t)$ can be expressed as

$$\alpha_i(t) = \sqrt{c_i^2(t) + H^2[c_i(t)]} \quad (2)$$

$$\theta_i(t) = \arctan\left(\frac{H[c_i(t)]}{c_i(t)}\right) \quad (3)$$

The instantaneous frequency $\omega_i(t)$ can be given by:

$$\omega_i(t) = \frac{d\theta_i(t)}{dt} \quad (4)$$

The frequency-time distribution of the amplitude is designated as the "Hilbert amplitude spectrum" $H(\omega, t)$, or simply "Hilbert spectrum".

$$H(\omega, t) = \text{Re} \sum_{i=1}^n \alpha_i(t) \exp(j \int \omega_i(t) dt) \quad (5)$$

Here, the residue $r_n(t)$ has been left out on purpose, for it is either a monotonic function or a constant. $\text{Re}\{\cdot\}$ denotes the real part of a complex quantity.

With the Hilbert Spectrum defined, the marginal spectrum $h(\omega)$ can be defined as

$$h(\omega) = \int_0^T H(\omega, t) dt \quad (6)$$

where T is the total data length. The marginal spectrum offers

a measure of the total amplitude contribution from each frequency value. If we just interest in the particular IMF which contend the frequency we just want investigate, local marginal spectrum can be used, and it is given as

$$h_i(\omega) = \int_0^T H_i(\omega, t) dt \quad (7)$$

The local marginal spectrum $h_i(\omega)$ offers a measure of the total amplitude contribution from the frequency that we are interested in.

The instantaneous energy $IE(t)$ provides information about the time variation of energy, which can be defined as

$$IE(t) = \int_{\omega_1}^{\omega_2} H^2(\omega, t) d\omega \quad (8)$$

We can use equation (9) to compute the local instantaneous energy $IE_i(t)$, which represent the energy variation produced by a single oscillation source which we are investigated.

$$IE_i(t) = \int_{\omega_1}^{\omega_2} H^2_i(\omega, t) d\omega \quad (9)$$

The combination of the empirical mode decomposition and the Hilbert spectral analysis is known as the ‘‘Hilbert-Huang transform’’.

B. BP neural network

In order to classify the EEG from different mental tasks, a three-layer BP neural network is structured using the features attracted by HHT. A sigmoid transfer function is applied between each layer. Levenberg-Marquardt method is applied to neural network training.

C. Data processing steps

1. Surface Laplacian filter was used to derivate of the instantaneous potential distribution. In this study, we investigated electrodes C3 and C4. The voltage of these two electrodes were calculated according to the following [9]:

$$V_j^{Lap} = V_j - \frac{1}{4} \sum_{k \in S_j} V_k \quad (10)$$

where V_j is the potential of the j^{th} electrode (here is C3, C4), and S_j is an index set of the four surrounding electrodes (here are C1, C5, CP3, FC3, C2, C4, CP4, FC4).

2. For each trail, EMD method was used, and the two channels C3 and C4 were decomposed into several IMFs and two residues. The IMFs which contains the frequency we interested in such as alpha band or beta band were selected in order to do further study. Fig. 2 displays the empirical mode decomposition in several IMFs of the raw C3 signal. In this case, IMF1 and IMF2 were selected to investigate.

3. Local marginal spectrum was calculated using the data recorded from 3s to 5.5s at electrode C3, C4 and the frequency band which energy difference most in three tasks was selected. According equation (9), local instantaneous energies at the specific frequency band were calculated as the features of different mental task.

4. The BP neural network as a classifier was trained by

the features extracted from signal using HHT. Classification accuracy was gained when the trained classifier was used to classify the test dataset.

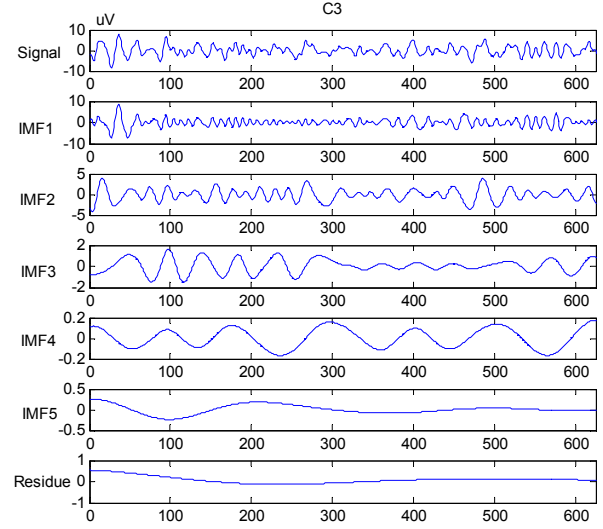


Fig. 2 IMFs of the signal recorded from electrode C3

IV. RESULTS

The data about each subject consist of 240 trails, including 80 repetitions of each type of mental tasks (left or right hand imagination or foot imagination). Ten fold cross validation is applied to test the performance of proposed procedure. The particular frequency band and average accuracy of each subject is shown in Table I.

Subject	Frequency band	Left hand	Right hand	Foot
Subject 1	8-14Hz	81.6	82.1	93.8
Subject 2	9-15Hz	78.3	81.4	85.6
Subject 3	10-15Hz	83.3	77.5	86.1

V. DISCUSSION

The Table I show that the classification accuracy of foot imagination is much higher than the other two motor imagery tasks (left or right hand imagination), especially in subject 1. The average marginal spectrums of subject 1 during each imagery task at electrode C3 and C4 are shown in Fig.3.

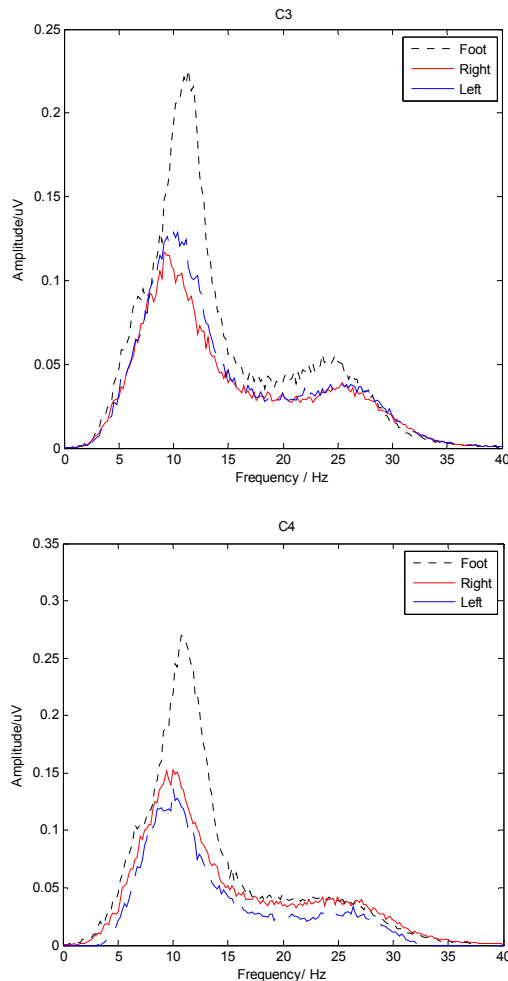


Fig. 3. Average marginal hilbert spectrum of subject 1 at electrode C3, C4

Fig.3. shows that the amplitude of mu rhythm in foot imagery task is much higher than hand imagery tasks. This phenomenon indicates that the energy among 8-13Hz frequency band varies greatly, when the subject is imagining the hand movement and foot movement. According to the research of Pfurtscheller [10], the hand motor imagery induced a significant mu ERD in both hemispheres, whereas foot motor imagery revealed a significant ERS in hand area on the cortex. That is the reason why the higher amplitude emerges in mu frequency band during the foot imagery, and also the reason why there is less difference in amplitude when left and right imagery happens. Another opinion presented by Pfurtscheller is lower mu components can be blocked or desynchronized by hand motor imagery and higher mu components can be enhanced by foot motor imagery. This phenomenon above can also be found in our study. During the hands imagery tasks, the center frequencies of mu rhythm are lower than foot. This fact can provide an alternative features for classification.

Previous researches of motor imagery tasks mainly focus on imagination of left hand and right hand movements. Although it is a normal way to classify EEG signals recorded during these two kinds of task according to the different level

of ERD. But, if it is necessary for a BCI system to perform an on/off control, it should be more effective to choose foot movement and hand movement as the mental tasks. And it is much easier to differentiate the signals between foot and hand movement tasks than between left hand and right hand movement tasks.

VI. CONCLUSION

In this paper, HHT is applied to analyze EEG signal in three different sorts of motor imagery tasks. The experiments based on imaginations of left hand, right hand and foot movements are performed. For each task, the energies of special frequency band at electrodes C3 and C4 are calculated as the features. A three layers neural network is structured to classify the signals which are recorded during different tasks. The classify accuracy shows that HHT is a very efficient method to extract the features from EEG signals on BCI research work. The result of the classification proves that a much higher classify accuracy can be reached with foot and hand imagery tasks. Furthermore, this conclusion will be very meaningful for a high efficiency BCI system design in future research.

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